

Sang Hyuk Hong

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EDUCATION

Korea University

B.S. in Physics

GPA: 4.09 / 4.50 (Major GPA: 4.19 / 4.50)

March 2019 – February 2023

Seoul, Republic of Korea

PUBLICATIONS

- [1] Laser mode-hopping assisted all-optical single beam pulsed atomic magnetometer with a sub-pT/ $\sqrt{\text{Hz}}$ sensitivity
J. H. Yoon, **S. H. Hong**, T. Jeong, S. H. Yim, K. M. Shim, and S. Lee
Under review at Physical Review A (2026) [Link]
- [2] Development and characterization of atom spin gyroscopes
S. H. Hong, S. Lee, S. H. Yim, S. R. Yeo, S. Lee, H. Yu, T. Jeong, J. B. Nam, D. Kim, and Y. Lim
Manuscript in preparation
- [3] Fabrication of an Rb–Xe gas cell with optical components attached by optical contact bonding for an atom spin gyroscope
S. H. Hong, S. H. Yim, T. Jeong, J. B. Nam, and S. Lee
Applied Optics (2025) [Link]
- [4] Aging test of an atomic vapor cell with Al₂O₃ wall coating on cubic glass
H. Kim, T. Jeong, **S. H. Hong**, J. B. Nam, S. Lee, Y. Lim, and S. H. Yim
Applied Optics (2025) [Link]
- [5] Wire-sealed atomic gas cells for noble-gas NMR applications
S. H. Hong, Y. Lim, S. H. Yim, S. Lee, H. Kim, T. Jeong, and J. B. Nam
Under review at Review of Scientific Instruments (2026)
- [6] Spin relaxation and frequency shift from time-periodic magnetic-field inhomogeneities in diffusing spin-1/2 gases
S. H. Hong and S. Lee
Manuscript in preparation

HONORS & SCHOLARSHIPS

National Scholarship for Study Overseas

2027 – 2029

National Institute for International Education

- Secured funding of up to US\$40,000 per year for the first two years of graduate study.

KFAS Study Abroad Workshop

2026

Korea Foundation for Advanced Studies (KFAS)

- Selected as one of 45 participants nationwide.

Commendation from the Minister of DAPA

2026

Defense Acquisition Program Administration (DAPA)

- Awarded for outstanding research achievements during military service.

National Scholarship for Natural Science and Engineering

2021 – 2022

Korea Student Aid Foundation

- Full tuition coverage plus stipends totaling US\$20,000.

RESEARCH EXPERIENCE

Seoul National University

Research Assistant

July 2026 – Present

Seoul, Republic of Korea

- Engineered coil and magnet systems for 2D and 3D magneto-optical traps (MOTs) for a dual-species Rb–Yb tweezer array platform.
- Developing an ultra-low-expansion (ULE) cavity system to stabilize lasers at 780, 399, and 556 nm for a dual-species Rb–Yb Bose–Einstein condensation experiment.

Agency for Defense Development

Research Officer for National Defense

June 2023 – May 2026

Daejeon, Republic of Korea

- Demonstrated single-beam all-optical scalar optically pumped magnetometers (OPMs) [1] for applications in vector magnetometry, magnetic navigation, magnetic anomaly detection, and magnetic field imaging.
- Realized navigation-grade low-SWaP (size, weight, and power) nuclear magnetic resonance gyroscopes (NMRGs) with bias stability of 0.01 deg/h [2], enabling in-house inertial measurement unit (IMU) and inertial navigation system (INS) development.
- Established novel fabrication techniques for atomic gas cells, enabling atomic sensors with compact form factors [3], prolonged cell lifetimes [4], and improved geometric symmetry and pressure reproducibility [5].
- Formulated theoretical and numerical frameworks for spin relaxation and frequency shifts induced by time-periodic magnetic-field inhomogeneities [6].

PATENTS

- **Heated-wire sealing method for atomic vapor cells**
KR Patent pending, 2025.
- **Vector magnetic-field measurement method using AC Stark effect in a scalar optically pumped magnetometer**
KR Patent pending, 2026.

SELECTED PRESENTATIONS

55th APS DAMOP Meeting

Poster Presentation

June 2024

Fort Worth, Texas, USA

- Analysis of heading errors in pulsed-mode and continuous-mode all-optical atomic magnetometers.

16th CLEO-PR

Poster Presentation

August 2024

Incheon, Republic of Korea

- Atom spin gyroscope using an Rb–Xe gas cell.

2025 Korean Physical Society Fall Meeting

Oral Presentation

October 2025

Gwangju, Republic of Korea

- AC magnetic-field-gradient-induced transverse relaxation for diffusive spin-1/2 particles in a cubic cell.

TECHNICAL SKILLS

Scientific Computing Python, Wolfram Mathematica.

Scientific Software AutoCAD, COMSOL Multiphysics, LabVIEW, Origin, SolidWorks.